

Teacher guide

Use the 50-page student workbook with the companion Practice and Activities booklet. The workbook contains illustrated teaching, all 18 extended responses and eight folio stages. The companion preserves all 180 course multiple-choice questions and 18 applied activities, with printable stimuli and working fields.

A classroom sequence

Teach the section reading and discuss its example. Students then complete the numbered knowledge check and applied activity in the companion. Use feedback to repair misconceptions before the extended response. Ask students to keep one clear piece of evidence for the relevant folio stage. Section numbers are consistent across the package.

Use the answers as reasoning support

Each section page below gives the correct choice and rationale for all ten MCQs, then expected evidence for the long response and applied activity. Accept alternative examples when they fit the source evidence and justify the relationship. The guide is formative teaching support; it does not add a formal marking scheme.

Practical and assessment authority

Before food production, confirm the current approved recipe, ingredients, equipment, food-allergy arrangements, room procedures, risk controls and supervision. Confirm the current HelloEats notification, due date, weighting, submission route and marking authority. Website or workbook completion is not proof of formal submission.

Sensitive and inclusive teaching

Use fictional cases to discuss food access and health. Avoid moral labels, body judgements or requests for private medical disclosure. Nutritional guidance is population guidance; diagnosed conditions and individual treatment remain with qualified professionals and the student's current management plan.

Print the core workbook at actual size on A4: 50 pages / 25 duplex sheets, long-edge binding. Print the companion separately. Word files retain native text, tables and response fields. Recheck pagination if editing changes text or fonts.

The current course repository is Year-9-Food-Selection-and-Health, revision c5e33e886b2ccdbf79cceac06f4cd6957240d247. This package resolves the old course URL and identifies the small source ambiguities in the source notes at the end.

ANSWERS / SECTION 1.1

Personal hygiene and readiness

Student reading page 3; extended response page 4. Companion section 1.1.

Knowledge check answers

1. C — The phone may transfer contamination back to clean hands. Frequently handled items can recontaminate washed hands.
2. A — Use soap and running water, clean all surfaces, then dry thoroughly. Soap, friction, rinsing and drying reduce material and transfer.
3. B — It creates a cleaner barrier between everyday clothes and food. The barrier reduces transfer from ordinary clothing.
4. D — It reduces hair falling into food and repeated hand-to-hair contact. It controls a physical hazard and a transfer route.
5. A — Tell the teacher and follow the setting's wound-protection directions. Reporting allows a suitable control or role to be chosen.
6. C — Some illnesses can spread harmful microorganisms through food. Early reporting supports a safe role decision.
7. B — They can become contaminated and need clean hands and task-aware changes. Gloves are one part of a hygiene system.
8. D — They can trap soil, hinder washing or become physical hazards. Their surfaces and shape can weaken hygiene controls.
9. B — Report health issues, prepare yourself, check the station, then review the method. This person-to-place order controls hazards before work.
10. C — Appearance alone cannot confirm food safety; use and cleaning status matter. Visible cleanliness is weaker than evidence of control.

Extended response

Accept a justified five-point routine covering reporting, hands, hair/clothing, personal items and station/method checks. A model chain is phone → hand → ready-to-eat bread: put the phone away, wash with soap and running water, then dry before handling food. Hair restraint controls falling hair and repeated touching; a clean apron reduces transfer from everyday clothing. Report symptoms and wounds so the teacher can choose controls or an alternative role. Check actual cleaning status rather than a shiny appearance, and keep clean equipment protected. Strong responses explain a pathway for every action, acknowledge that gloves can become contaminated and name one local procedure to confirm. The labelled chain and room-procedure question belong in the folio evidence.

Response quality criteria

- Uses cause-and-effect reasoning rather than a rule list.
- Covers personal preparation and the work area.
- Distinguishes visible dirt from invisible contamination.
- Does not invent school-specific procedures.

ANSWERS / SECTION 1.2

Contamination, temperature and food safety

Student reading page 5; extended response page 6. Companion section 1.2.

Knowledge check answers

1. B — Harmful bacteria multiplying in poorly chilled cooked food. Harmful bacteria are biological hazards.
2. D — A raw-chicken knife is used on salad without cleaning. The knife carries a hazard to ready-to-eat food.
3. A — It must also be clean, correctly assigned and separated. Separation works only with cleaning and workflow.
4. C — Cold at 5°C or colder; hot at 60°C or hotter. These are the standard control boundaries.
5. B — Use it immediately and do not refrigerate it for later. Two to four hours means use immediately.
6. D — Discard it. More than four hours requires disposal.
7. A — It gives a measurement to compare with a boundary. Measured evidence supports the decision.
8. C — Clean away soil before sanitising as directed. Sanitising works on an effectively cleaned surface.
9. B — Cooking may destroy many microbes, but food can be contaminated again. Post-cooking handling can reintroduce hazards.
10. D — Verify ingredients and use effective separation, cleaning and controlled utensils. The controls address source and transfer.

Extended response

Look for four plausible hazards and explicit routes, such as raw-food microbes moving by knife to salad, allergen residue transferring on shared utensils, a cleaning chemical near food and packaging fragments entering the filling. Students should place controls at the transfer points: separate storage/workflow, cleaned equipment, effective allergen controls and safe chemical storage. The temperature decision must include cumulative time and a measured holding boundary. For a supplied three-hour history, use immediately and do not chill for later; for more than four hours, discard. Cleaning must precede sanitising under approved directions. An excellent plan distinguishes observed hazards from possible ones, names missing time information and records a separate defensible decision for each food without assuming cooking fixes every hazard.

Response quality criteria

- Applies hazard categories accurately.
- Explains transfer and growth mechanisms.
- Uses Australian temperature and time guidance correctly.
- Combines controls rather than claiming one guarantee.

ANSWERS / SECTION 1.3

Safe decisions and incident response

Student reading page 7; extended response page 8. Companion section 1.3.

Knowledge check answers

1. C — A hazard can cause harm; risk considers likelihood and consequence. This distinction guides proportionate controls.
2. A — Its edge is unseen and a hand may contact it unexpectedly. Poor visibility increases likelihood of harm.
3. D — Stop, keep others clear where safe, and tell the teacher. This prevents escalation and brings authorised help.
4. B — Damaged lead at Station 4; appliance untouched; everyone clear. It states location, hazard and current control.
5. C — An unsuitable action can create a second hazard. Special incidents need matched, authorised controls.
6. A — Isolate the area and treat the food as potentially contaminated. Small fragments can be difficult to detect.
7. D — Keep routes clear and announce movement with hot or sharp items. Layout and communication reduce unexpected contact.
8. B — It reveals a control weakness before harm occurs. Near misses supply prevention evidence.
9. C — Water covered the floor between sink and oven. This reports a visible condition.
10. A — Place tasks so hot, sharp and wet pathways do not cross. Layout reduces exposure before work.

Extended response

Accept any realistic near miss with separate observations, plausible harm and justified likelihood/consequence. For a wet route between sink and oven, the observed water is the hazard source; someone carrying a hot item could slip and be burned. Stop movement, keep classmates clear where safe and report the exact location and condition. Cleanup, first aid or equipment action follows teacher directions; do not reward improvised electrical or fire responses. A before-and-after workflow could move wet tasks away from hot-item routes and establish a clear landing zone. A success measure might be an unobstructed route check before each transfer. Strong work connects the proposed change to the pathway, avoids blame and explains why reporting a near miss prevents future harm.

Response quality criteria

- Separates fact from inference and avoids blame.
- Uses hazard, risk and control accurately.
- Keeps actions within a student's role.
- Links redesign to the original pathway.

ANSWERS / SECTION 2.1

Six nutrients and their roles

Student reading page 9; extended response page 10. Companion section 2.1.

Knowledge check answers

1. B — Carbohydrate, protein and fat. These are the energy-yielding nutrients in the six-group model.
2. D — Most foods contain combinations of nutrients and components. Real foods make overlapping contributions.
3. A — It is broken down mainly to glucose that can fuel cells. Glucose is an important physical and mental fuel.
4. C — Fibre, micronutrients, structure and nutrient profiles differ. The source brings different accompanying components.
5. B — It supplies amino acids for tissues, enzymes and other components. Protein supports structure and functional molecules.
6. D — Chickpeas. Legumes provide protein and other nutrients.
7. A — Fat has essential roles, while type, amount and source affect the choice. Fat supports cells, absorption and energy; sources differ.
8. C — Iron—oxygen transport; calcium—bones and teeth. These are established mineral functions.
9. B — It supports transport, temperature control and reactions. Its regulatory roles are essential.
10. D — Different foods provide complementary nutrients and fibre. Complementary contributions make the pattern robust.

Extended response

Accept a varied lunch with at least four foods and an accurate three-function map. For example, a wholegrain roll, chickpea hummus, carrot, plain yoghurt and water can contribute carbohydrate for energy, protein for tissue repair, minerals such as calcium, fibre and fluid. Students need not calculate exact intake or claim that the example meets a person's entire daily requirements. Look for foods placed in more than one category where justified, correct distinction between nutrient and food group, and a reason for complementarity. A strong superfood evaluation explains that no single item provides every useful nutrient/component or replaces a varied eating pattern. Allergy, culture, affordability and approved practical conditions remain checks rather than assumed facts.

Response quality criteria

- Uses all six nutrient groups accurately.
- Includes fibre without mislabelling it in the six-group model.
- Explains functions through cause and effect.
- Avoids moral good/bad labels.

ANSWERS / SECTION 2.2

Energy, growth and repair

Student reading page 11; extended response page 12. Companion section 2.2.

Knowledge check answers

1. C — Energy. Kilojoules are energy units.
2. A — Breathing, circulation and temperature control. Basic functions need continuous energy.
3. D — Balance reflects changing intake and use over time. Patterns and context are required.
4. B — Growth and physical activity. Both affect energy use.
5. A — Useful nutrient contribution alongside kilojoules. It broadens evidence beyond energy.
6. C — Compare nutrients, ingredients, portion and purpose. Similar energy does not mean similar contribution.
7. D — Carbohydrate food, protein food, plant foods and fluid. The parts make complementary contributions.
8. B — It standardises amount when serving sizes differ. A common basis makes comparison fairer.
9. A — When estimating the planned portion's contribution. It connects data to realistic intake.
10. C — This option supplies more energy and may fit longer training. It connects measured energy to context.

Extended response

For Alex's supplied sport context, a reasonable conditional preference is Option A because the wholegrain chicken-and-salad wrap, plain yoghurt, banana and water offer complementary carbohydrate, protein, plant foods and fluid. This is a qualitative judgement: exact serves and label data are absent, so energy or nutrient totals must not be invented. Option B also contributes foods and nutrients and may suit a different portion, availability or preference. Strong responses compare both fairly, explain the time between lunch and sport and note that a cool pack supports storage planning without proving actual temperature control. Require at least one condition that could change the choice, such as allergy, culture, appetite, budget or measured portion information. Do not prescribe personal calorie targets.

Response quality criteria

- Keeps kilojoules and nutrient density distinct.
- Connects choice to growth, activity and repair.
- Considers portion and pattern.
- Uses neutral health language.

ANSWERS / SECTION 2.3

Digestion and nutrient absorption

Student reading page 13; extended response page 14. Companion section 2.3.

Knowledge check answers

1. B — Digestion breaks food down; absorption moves suitable molecules across the gut wall. Preparation and transfer are distinct.
2. D — It increases surface area available to enzymes. Smaller pieces improve enzyme contact.
3. A — Peristaltic muscular contractions. Peristalsis moves material through the tract.
4. C — Muscular mixing with acidic juice and early protein digestion. These are stomach functions.
5. B — Small intestine. It is specialised for digestion and absorption.
6. D — It disperses fat droplets to increase enzyme access. Emulsification increases surface area.
7. A — Large area, thin surfaces and nearby transport vessels support transfer. Structure is suited to absorption.
8. C — Simple sugars and amino acids. These are small absorbable products.
9. B — Absorb water and form material for elimination. These are key functions.
10. D — It supports bowel function and interacts with gut microorganisms. These are recognised roles.

Extended response

A strong mixed-meal pathway follows mouth → oesophagus → stomach → small intestine → large intestine. Chewing and stomach mixing are mechanical; enzyme action breaks molecules chemically. Salivary enzymes start some carbohydrate digestion, stomach enzymes begin substantial protein digestion, and pancreatic/intestinal enzymes continue digestion of the three macronutrients. Bile comes from the liver via storage in the gall bladder and disperses fat droplets without being an enzyme. Most absorption occurs across small-intestine villi; simple sugars and amino acids enter blood and many fat products enter lymph. The large intestine absorbs water and forms material for elimination. Credit explanations of surface area, thin walls and nearby vessels. Reject the stomach-only claim using this distributed sequence, rather than merely saying it is wrong.

Response quality criteria

- Distinguishes digestion, absorption and transport.
- Explains mechanisms rather than listing organs.
- Connects surface area to chewing and villi.
- Includes water and fibre.

ANSWERS / SECTION 3.1

Ingredients, servings and nutrition panels

Student reading page 15; extended response page 16. Companion section 3.1.

Knowledge check answers

1. B — Descending ingoing weight when made. Order reflects formulation by weight.
2. D — The exact amount of every ingredient. Exact quantities are not supplied for all ingredients.
3. A — Length does not reveal each ingredient's amount, role or effect. Interpretation needs composition and context.
4. C — The amount used to calculate per-serve values. It is the panel's calculation basis.
5. A — Different stated serves can distort the apparent comparison. Equal-looking serves may represent unequal amounts.
6. D — Per 100 g. A standard amount supports fair comparison.
7. B — Naturally present plus added sugars where present. This is why ingredients add context.
8. C — Do not eat it after the use-by date. Use-by concerns safety.
9. A — Expected quality, not an automatic safety cut-off. Best-before is primarily about quality.
10. D — Current ingredients and allergen information used with the management plan. Current safety evidence and the person's plan control the decision.

Extended response

An effective label guide begins with ingredients/allergens, date and storage where safety is relevant, then compares similar snacks using the same 100 g or 100 mL basis. It checks each serve's mass and serves per pack before translating data to the realistic portion. Explain that ingredient order is descending ingoing weight, not a list of exact percentages, and that total sugars may combine natural and added sources. Use-by concerns safety; best-before mainly concerns quality under correct storage. Accept different scanning order if the stated purpose and safety priorities are justified. The annotated package should connect each question to the correct evidence zone. Reward explicit limits, such as "no added sugar does not mean sugar-free" and "a favourable panel cannot establish allergy suitability".

Response quality criteria

- Uses Australian label terminology.
- Separates safety, formulation, comparison and portion evidence.
- Explains limitations rather than treating the panel as a verdict.
- Prioritises allergens and use-by information.

ANSWERS / SECTION 3.2

Comparing foods fairly

Student reading page 17; extended response page 18. Companion section 3.2.

Knowledge check answers

1. C — Audience, occasion and purpose. Purpose determines relevant evidence.
2. A — They serve a similar role, so differences answer the question. Like-with-like improves relevance.
3. D — Per 100 g. A common amount supports fairness.
4. B — It determines the amount actually contributed. Quantity translates baseline to the meal.
5. C — Compare per 100 g and the same realistic portion. This tests whether the advantage is real.
6. A — Relevant nutrients, portion, food-group contribution and context. They connect evidence to purpose.
7. D — Total combines sources; ingredients add source context. Using both prevents overstatement.
8. B — Serious allergy or no cold storage. Safety and access can control usability.
9. C — For this refrigerated snack and portion, B fits better; allergy needs would change it. It states context, evidence and a condition.
10. A — State uncertainty and compare a range or identify what to confirm. Transparent assumptions preserve limits.

Extended response

For a portable refrigerated lunch, a defensible recommendation may favour fictional yoghurt A for its higher protein and lower total sugars per 100 g, provided milk is suitable and refrigeration is maintained. Show both baselines and tub amounts: A's whole tub supplies 510 kJ, 11.9 g protein, 8.5 g sugars and 85 mg sodium; B supplies 500 kJ, 5.6 g protein, 16.3 g sugars and 75 mg sodium. A is lower in sodium per 100 g yet higher per tub because its tub is larger. This is a useful trade-off, not a contradiction. Accept B if evidence and context justify it. Price, preference and confirmed allergy needs are absent. Rounded calculations should be identified rather than marked as false precision.

Response quality criteria

- Uses a common amount accurately.
- Selects criteria for the purpose.
- Shows trade-offs and uncertainty.
- Avoids universal health claims.

ANSWERS / SECTION 3.3

Marketing claims, ethics and health

Student reading page 19; extended response page 20. Companion section 3.3.

Knowledge check answers

1. B — It may highlight one benefit while excluding other relevant evidence. Selection and omission shape interpretation.
2. D — It meets the conditions for that claim, but other qualities need checking. The conclusion stays within the claim.
3. A — Match exact wording to relevant label evidence and do not extend it. Claim and evidence should align.
4. C — To compare similar packaged foods within a category. This is the intended comparison.
5. B — The star does not replace formulation and allergen evidence. The sources answer different questions.
6. D — A popular athlete enjoying the product. Endorsement suggests admired people choose it.
7. A — What information would change the impression if shown beside the claim? This tests missing context.
8. C — The food's nutrient profile suits the eater. Environmental and nutrition evidence are separate.
9. B — Check evidence, identify omissions and acknowledge real benefits. Critical evaluation recognises strengths and limits.
10. D — State a conditional judgement and name what is needed. Transparent uncertainty is responsible evaluation.

Extended response

Use the supplied Berry Crunch promotion. “Made with real berries” is supported by berry purée at 2%, but the proportion limits any impression that berries dominate. “Natural energy” is undefined, while “Fuel your best” lacks a stated performance measure. The ingredients are led by oats (42%), yoghurt-style coating (26%, including sugar/oil/milk powder) and sugar. Students should not infer exact nutrient totals from this list. Omissions include portion, nutrition panel, total price, allergen relevance and evidence for environmental claims. Strong ethical evaluation recognises both the actual berry ingredient and the risk of exaggerated impressions. A fairer message might identify an oat-based snack containing 2% berry purée, give appropriate allergen information and direct consumers to complete evidence without health guarantees.

Response quality criteria

- Distinguishes persuasion, regulated information and inference.
- Uses Health Star Rating appropriately.
- Explains audience impact without assuming students are powerless.
- Makes uncertainty and trade-offs explicit.

ANSWERS / SECTION 4.1

Active non-nutrients and food variety

Student reading page 21; extended response page 22. Companion section 4.1.

Knowledge check answers

1. A — A food component that can affect body processes but is not one of the six nutrient groups Fibre and plant compounds can support functions alongside the six nutrient groups. Water remains a nutrient; the original source feedback grouped it incorrectly.
2. B — It can add bulk or be fermented after resisting full digestion in the small intestine Correct. This follows the mechanism in Fibre works through the digestive system.
3. A — Add several varied fibre-rich foods gradually and drink water Correct. Gradual variety with water reflects the mechanism and avoids an all-or-nothing response.
4. C — A range of plant colours can help broaden the mix of food components Correct. Colour is a useful variety prompt when the claim remains modest and pattern-based.
5. A — It is mainly a marketing term and can exaggerate one food's role Correct. The label can distract from the evidence about the whole eating pattern.
6. B — They add variety, texture and a broader mix of food components Correct. This is a specific, proportionate explanation of the redesign.
7. A — Whether the person likes and can access the foods Correct. Preference and access influence whether the plan can be followed in real life.
8. C — Both can support normal digestive function and should be considered together Correct. The section explains their connected role without claiming they are the same.
9. B — This option adds legumes and vegetables, increasing variety and likely fibre Correct. The claim identifies observable ingredients and a reasonable mechanism.
10. D — A varied pattern across time is more useful than ranking single foods as good or bad Correct. This is the section's central evidence-based, non-moralising idea.

Extended response

Accept any familiar lunch assessed respectfully, followed by two realistic changes tied to a purpose. A white-bread-and-cheese sandwich with water could use wholegrain bread for additional fibre and add tomato or carrot for plant variety and texture. The response should recognise existing contributions, then explain how each change broadens useful components. Do not require an expensive ingredient, a fixed colour quota or a complete menu replacement. Cautious statements include “contributes fibre” or “broadens the food-group mix”; reject detox, cure and guaranteed-prevention claims. A strong evaluation checks affordability, taste, access, allergies and storage. It also distinguishes water as a nutrient from fibre and plant compounds considered beyond the six-group model. The before-and-after comparison needs two evidence statements.

Response quality criteria

- Uses accurate fibre, water and variety concepts
- Links each redesign decision to evidence and context
- Avoids moral labels and unsupported health promises
- Explains both benefits and limits

ANSWERS / SECTION 4.2

Nutrition across the lifecycle

Student reading page 23; extended response page 24. Companion section 4.2.

Knowledge check answers

1. B — Individual activity, health, access and preferences also affect needs Correct. Life stage interacts with several personal and environmental factors.
2. A — Growth and level of activity Correct. Growth and activity are relevant physiological and lifestyle factors.
3. C — Some may need nutrient-dense or easier-to-handle foods, but the brief must identify the need Correct. It recognises possible design factors without assuming they apply to everyone.
4. C — Support broad comparison and planning while recognising individual variation Correct. This matches the section's evidence boundary.
5. A — Choose a practical varied option, explain how it fits the schedule and check storage Correct. It connects a stated need to food features and practical checks.
6. D — Two people of the same age may need different designs because their contexts differ Correct. It uses age as one factor and keeps the person visible.
7. C — An unstated swallowing difficulty Correct. A swallowing difficulty is significant and must not be inferred from age or appearance.
8. A — Body processes, appetite or ability to prepare and eat food may change Correct. Several physiological and practical factors may change, so context must be confirmed.
9. B — Read brief, identify need, select feature, explain response, check practicality Correct. It is the complete reasoned chain described in the theory.
10. A — Does it respond to the stated need and remain acceptable, safe and practical? Correct. This checks both the evidence-based purpose and real-world feasibility.

Extended response

Compare Jordan and Ari using only the supplied facts. Jordan is 14, has ten minutes for lunch, an insulated bag, sport after school and a preference for familiar savoury food. A compact, easy-to-eat wrap may fit the short break; check portion and actual cold storage. Ari is 72, eats at home, prefers softer texture and an easy-to-hold serve, and enjoys family meals. A softer presentation and manageable serving format can respond to those stated needs without diagnosing swallowing difficulty. Both cases lack allergy, cultural and medical requirements; Jordan also lacks budget information. Strong work highlights these unknowns and explains at least one adaptation per person through need → feature → reason → practical check. Other evidence-based concepts are valid.

Response quality criteria

- Separates life-stage guidance from individual evidence
- Uses accurate cause-and-effect reasoning
- Avoids diagnosis, prescription and stereotypes
- Checks practicality, preference and safety

ANSWERS / SECTION 4.3

Personal, social, cultural and digital influences

Student reading page 25; extended response page 26. Companion section 4.3.

Knowledge check answers

1. A — Many interacting personal and environmental factors shape it Correct. The systems view traces interacting influences rather than blaming one trait.
2. A — A person's sensory preferences and allergies Correct. Preference and allergy information relate directly to the individual.
3. C — Ask what the food means to the user and analyse its actual ingredients and context Correct. This combines respectful inquiry with specific food evidence.
4. A — Which options appear first and how discounts are framed Correct. Interface order and promotions can shape attention and convenience.
5. A — It may indicate a commercial interest in the audience buying the product Correct. A commercial relationship is relevant context when assessing persuasion.
6. B — Name the purpose and influences acting on the choice Correct. Naming purpose and pressure makes later evidence checking relevant.
7. B — The ingredient list, nutrition panel and source of the claim Correct. These sources allow the product and message to be checked directly.
8. C — Consider serve, additions, frequency, cost, enjoyment and the person's overall context Correct. This is a balanced, contextual analysis.
9. B — Their needs, preferences, culture, access or plans may differ Correct. Multiple legitimate factors may explain different choices.
10. C — Identify pressures, use available evidence and make the most workable decision for now Correct. It combines agency, evidence and realistic conditions.

Extended response

A strong response explains at least three interacting influences rather than only naming categories. For example, thirst and taste may interact with a friend's suggestion, a ten-minute discount and a limited budget. A platform's placement or influencer message can make one option more visible without proving it is suitable. The pause stage names purpose and pressures; the check stage seeks ingredients, nutrition information, total price and reliable support for a claim; the decide stage states a workable option and its limits. Avoid blaming, moral labels and cultural assumptions. An equally defensible alternative may follow different preferences or constraints. The folio influence map should show arrows explaining reinforcement or conflict, and the final reflection should identify at least one piece of missing information.

Response quality criteria

- Explains interaction rather than listing influences
- Uses evidence beyond popularity or appearance
- Respects culture, preference and access
- Avoids blame, shame and absolute claims

ANSWERS / SECTION 5.1

Australian food-consumption patterns

Student reading page 27; extended response page 28. Companion section 5.1.

Knowledge check answers

1. A — A repeated tendency in a defined group, time or setting Correct. A pattern summarises defined evidence and allows for variation.
2. A — Different measures such as purchases and food eaten support different conclusions Correct. Claims must match the evidence actually collected.
3. B — That the household purchased six apples in the recorded period Correct. This claim stays within the evidence.
4. B — The surveyed group averaged the reported amount, with individual variation Correct. It states the group finding and its key limit.
5. A — Food access, culture, age or income Correct. These factors can shape real opportunities and preferences.
6. A — Whether measures and methods are sufficiently comparable Correct. A change in method can create an apparent difference unrelated to behaviour.
7. A — As a comparison guide that can reveal design opportunities Correct. Guidance helps interpret opportunities while context remains important.
8. A — Test an appealing affordable vegetable-rich option with the intended students Correct. The response is targeted, testable and matched to the setting.
9. A — The intended audience, barrier, intended change and success measure Correct. These elements connect the response to evidence and evaluation.
10. B — Population patterns can guide action when their scope and limits are clear Correct. This captures both the value and boundary of population evidence.

Extended response

Require accurate identification of the current ABS excerpt: children aged 2–17, 2023 usual-consumption estimates, 2.1% meeting the vegetable/legume/bean recommendation and 50.7% meeting fruit, released 10 July 2026. The response must distinguish these figures from data about this school or a reason for the pattern. A proportionate design proposal might test an affordable vegetable-rich lunch component with a defined group, after investigating access, price and sensory preferences. Its success measure could be structured acceptability feedback or a clearly specified trial measure, not a national health claim. Credit an explicit local evidence gap and caution about comparing unlike survey methods. Do not require invented school data, causal explanations or a diagnosis of the children represented.

Response quality criteria

- Claims stay within the measured evidence
- Averages and variation are interpreted accurately
- The response is targeted and proportionate
- Evaluation includes a meaningful success measure

ANSWERS / SECTION 5.2

Convenience, delivery and food technology

Student reading page 29; extended response page 30. Companion section 5.2.

Knowledge check answers

1. A — Freezing can extend useful storage while slowing microbial growth Correct. This identifies a specific preservation function and mechanism.
2. A — Heat and a sealed container Correct. Heat treatment and sealing are core parts of the preservation system.
3. A — Ingredients, allergen declarations, dates and storage directions Correct. These details help check suitability and handling.
4. A — It cannot show the full time-and-temperature history Correct. A seal can show possible tampering but not every condition during transport.
5. A — Receive it promptly and follow storage directions Correct. The customer's decisions form the final part of the control chain.
6. A — Through rankings, fees, bundles and promotions Correct. Interface and pricing choices affect visibility and perceived value.
7. A — Menu price plus delivery fee, service fee and any minimum order Correct. Total cost includes the charges needed to complete the purchase.
8. A — It may save planning and build skills but can add cost or packaging Correct. It identifies possible benefits and trade-offs that depend on context.
9. A — Safety, nutrition, cost, appeal, packaging and user needs Correct. A multi-criteria comparison reveals the whole trade-off.
10. B — Select or improve convenience responsibly for a stated context Correct. Responsible design balances the user's purpose with evidence and trade-offs.

Extended response

For Casey's stated context, the home-assembled rice bowl appears the stronger time fit: about 12 minutes, compared with a delivery estimate of 25–40 minutes that may exceed the 30-minute need. A microwave and fridge are available, and ingredient/storage information is supplied for the home option. This is conditional because budget, allergies and preferences still need confirmation; the estimates are not guaranteed measurements. Compare packaging explicitly: one can and two packs versus a single-use bowl and bag, without inventing lifecycle impacts. The delivery choice needs ingredient/allergen, portion and arrival-temperature evidence. Strong answers trace preparation through eating, identify an essential control point and suggest an improvement that retains convenience. Neither route is automatically safer or nutritionally superior simply because of its label.

Response quality criteria

- Uses a complete set of relevant criteria
- Explains preservation or delivery mechanisms accurately
- Recommendation responds to the stated user
- Trade-offs and one realistic improvement are explicit

ANSWERS / SECTION 5.3

Designing responsible food solutions

Student reading page 31; extended response page 32. Companion section 5.3.

Knowledge check answers

1. B — Develop a portable varied lunch within the stated budget and safe-storage conditions Correct. It defines audience-relevant criteria and boundaries.
2. A — A quality used to judge whether the solution succeeds Correct. Criteria make evaluation explicit and fair.
3. A — A maximum preparation time set by the brief Correct. The time limit bounds what can be planned and produced.
4. A — To compare different ways of meeting the same criteria Correct. Alternatives reveal trade-offs and prevent premature commitment.
5. B — The sealed reusable container supports portability and the cold-storage plan addresses the perishable filling Correct. It links specific features to criteria and a safety control.
6. A — Structured feedback from intended users Correct. Intended-user feedback directly tests sensory experience.
7. A — Individual packaging improves portability but may increase material use Correct. One feature helps a criterion while creating an environmental concern.
8. A — The package looks green, so its total impact must be low Correct. Colour and appearance do not establish lifecycle impact.
9. B — The cost met the limit, but two tasters found the filling dry, so the next trial will adjust moisture while keeping safe storage Correct. It reports evidence, a limitation and a targeted next step.
10. A — It supplies evidence for the next design decision Correct. Evaluation closes one cycle and begins informed improvement.

Extended response

For Jamie, state the constraints: cold school lunch at 1 pm, insulated bag available, ingredient-cost target at most \$5 and home assembly within 15 minutes. Allergies, culture and medical requirements remain unknown. Compare at least two concepts in the long response, while the linked activity compares three. A useful matrix includes food variety, practicality, appeal, portability and evidence for safe holding. Do not award invented cost or temperature results; identify the calculation, label or test needed. The chosen concept requires a reasoned trade-off and a test plan, such as checking containment or obtaining intended-user feedback. Safety and allergen suitability are prerequisites rather than low-weight preferences. Accept different concepts when the evidence and limits are transparent and no approved recipe is implied.

Response quality criteria

- Criteria and constraints are specific and traceable to the brief
- Decisions use nutrition, safety and user evidence
- Trade-offs are genuine and not hidden
- Testing and improvement are measurable

ANSWERS / SECTION 6.1

Under- and overnutrition

Student reading page 33; extended response page 34. Companion section 6.1.

Knowledge check answers

1. A — A mismatch between needs and the energy or nutrients received or used Correct. The definition includes undernutrition, overnutrition and imbalance.
2. A — Yes, energy and nutrient adequacy are related but not identical Correct. A narrow energy-dense pattern may still lack fibre, vitamins or minerals.
3. A — Nutrition effects and adequacy depend on repeated patterns and wider context Correct. One event does not establish the person's usual intake or needs.
4. A — Food insecurity, illness affecting absorption or limited preparation access Correct. These factors can restrict intake or the body's use of nutrients.
5. A — It can involve sustained excess or imbalance shaped by biological and environmental factors Correct. This recognises the mechanism and its interacting influences.
6. A — Encourage contact with a trusted adult or appropriate support Correct. This respects privacy and routes the concern to people who can help.
7. A — Improve access to affordable varied food, meal time and drinking water Correct. It changes practical conditions that shape repeated choices.
8. A — The cause may involve access, illness, absorption, safety or other barriers Correct. A useful response must match the pathway and context.
9. A — The recorded pattern contains limited variety across several days Correct. This stays with observable case evidence and avoids diagnosis.
10. A — Design and communicate from reliable guidance without diagnosing or prescribing Correct. This preserves a useful educational role and directs clinical concerns appropriately.

Extended response

For a fictional case, require evidence → possible mechanism → interacting context → respectful support. Morgan's repeated missed breakfasts and limited lunches alongside food shortage may mean inadequate energy or variety, but tiredness cannot establish a diagnosis. The household with frequent large takeaway serves and sweet drinks may face a pattern that merits further investigation; actual intake and needs remain unknown. Taylor's narrow variety could limit fibre and micronutrient contributions even if energy were adequate. Accept analyses of any supplied case when they distinguish observed facts from hypotheses. Responses should address access, time, affordability or other stated barriers and identify a private trusted-adult/professional pathway. Reject public body judgement, prescribed diets and guaranteed treatment. The mechanism map needs a clear boundary statement.

Response quality criteria

- Distinguishes energy, nutrients and pattern duration
- Uses cautious non-diagnostic language
- Explains interacting causes without blame
- Routes support safely and respects privacy

ANSWERS / SECTION 6.2

Nutrition-related conditions and prevention

Student reading page 35; extended response page 36. Companion section 6.2.

Knowledge check answers

1. A — It is associated with a greater chance of an outcome but does not guarantee it Correct. Risk describes probability and association, not certainty.
2. A — Tiredness has many causes and anaemia can have different causes that require testing Correct. Symptoms are non-specific and diagnosis requires health-professional assessment.
3. A — Iron is needed to make haemoglobin, but food advice cannot identify the cause of someone's symptoms Correct. It explains the mechanism and preserves the clinical boundary.
4. A — Blood pressure, smoking, family history, diabetes, movement and dietary patterns Correct. Cardiovascular risk develops through several interacting factors.
5. A — Calcium, vitamin D, protein, hormones and weight-bearing activity can all matter Correct. Bone health is supported through multiple nutrients and body processes across life.
6. A — The body does not use insulin effectively and may not make enough for intended glucose regulation Correct. This describes the core glucose-regulation mechanism in age-appropriate terms.
7. A — Study mechanisms and supportive environments without weighing or commenting on classmates' bodies Correct. It teaches the content while protecting dignity and privacy.
8. A — A varied eating pattern and supportive environment can reduce risk for some conditions Correct. Reduce risk reflects evidence without promising certainty.
9. A — Allergy, intolerance, vegan choices, culture and availability may affect suitability Correct. Inclusive design responds to real user needs and values.
10. A — Speak privately with a trusted adult and seek qualified health advice Correct. Personal diagnosis and management require appropriate support and privacy.

Extended response

Accept any condition taught in the section with a correct mechanism, interacting risk/protective factors and an inclusive food-design response. For anaemia, explain iron's role in haemoglobin and oxygen transport while recognising multiple causes and the need for testing. For cardiovascular health, link heart/blood-vessel function to several relevant factors rather than one nutrient. For bone health, include calcium alongside vitamin D, protein, activity and wider influences. For type 2 diabetes, explain ineffective insulin response and possible insufficient production. The proposed food option may support a varied pattern but cannot promise prevention or treatment. Require cautious wording, a modifiable and a non-modifiable/contextual factor, and a private professional-help boundary. No diagnosis from fatigue, photographs or body size is acceptable.

Response quality criteria

- Mechanism is accurate and age appropriate
- Risk language avoids certainty and blame
- Design response is inclusive and evidence based
- Clinical and privacy boundaries are explicit

ANSWERS / SECTION 6.3

Australian guidance and the HelloEats challenge

Student reading page 37; extended response page 38. Companion section 6.3.

Knowledge check answers

1. A — Support planning a varied eating pattern across food groups Correct. The guide supports broad pattern planning.
2. A — It supports consistent comparison between similar products Correct. The common basis reduces confusion from different serving sizes.
3. A — Ingredients in descending order by ingoing weight Correct. This helps identify what the product contains and supports allergen checking.
4. A — As one aid for comparing similar packaged foods alongside other evidence Correct. The rating is most useful within an appropriate category and with the full label.
5. A — Compare two similar products using the same per-100 measure, then check serves, ingredients and purpose Correct. It uses a consistent basis and adds context.
6. A — A formative food-design practice context Correct. This site supports formative practice; any formal conditions come only from the current teacher-issued notification.
7. A — Current due date, weighting, marks or submission method Correct. These formal conditions require a current authorised notification.
8. A — The formative draft is stored in that browser for revisit or export Correct. Local persistence supports learning but is not submission evidence.
9. B — The concept responds to the user, uses guidance and label evidence, includes safety confirmation and explains trade-offs Correct. It joins evidence, user fit, safety and responsible evaluation.
10. A — The teacher confirms the practical plan, safety controls, ingredients and assessment status Correct. These are teacher-controlled facts and processes outside the draft website brief.

Extended response

A strong HelloEats draft defines a fictional Year 9 user and occasion, states criteria/constraints and presents a ready-to-eat concept with reasons. AGHE should support variety, while the complete yoghurt data supplies a fair component comparison. For example, A's higher protein and lower sugars per 100 g may inform the choice, but both contain milk and require refrigeration; do not assume safety from a favourable panel. Include cost, packaging, appeal and access trade-offs with missing values explicitly identified. The source list should separate official guidance from fictional label data. The status statement must say formative learning evidence, not submitted. Current notification and teacher confirmation control formal conditions and any production; no invented recipe quantities, marks, dates or medical promises are acceptable.

Response quality criteria

- Uses each guidance and label tool for its correct purpose
- Design decisions respond to a clearly defined fictional user
- Safety, allergens and trade-offs are visible
- Draft, formative and formal-assessment states remain unmistakably separate
- No dates, marks, weighting or submission claims are invented

TEACHING NOTES / STAGES 1–4

Folio evidence and feedback

1. Safe-practice decisions

1. Safe-practice decisions. Expected evidence: Look for a clear context and hazard, an accurate source → route → harm explanation, and a control that interrupts that route. A student may describe a proposed action if no real practical occurred, but must not imply observed performance. Require a reference to the current teacher or room procedure. A labelled drawing can be evidence; it need not include a photograph of a learner.

2. Nutrients and the body

2. Nutrients and the body. Expected evidence: Expect one precise nutrient–function mechanism and a food example, such as protein supplying amino acids for repair or iron contributing to haemoglobin and oxygen transport. The explanation should distinguish nutrient from food and avoid claiming one item supplies every need. A cause-and-effect chain plus a clear limit is stronger than a list of nutrient names.

3. Food-label comparison

3. Food-label comparison. Expected evidence: Expect a named audience and occasion, matched per-100-g or per-100-mL comparison, realistic portions and relevant ingredients/allergen context. The learner should show one trade-off and a conditional recommendation. The complete fictional yoghurt data is an acceptable evidence source if labelled fictional. Missing price or personal requirements must be stated rather than guessed.

4. Influences audit

4. Influences audit. Expected evidence: Expect at least three influences and one explained interaction, with respectful language about culture, access and preference. A digital claim should be checked for source, commercial purpose, evidence and omissions. The proposed response must fit the actual constraints. No particular food purchase is the single correct outcome; assess the reasoning and limits.

TEACHING NOTES / STAGES 5–8

Folio evidence and feedback

5. Patterns and technology

5. Patterns and technology. Expected evidence: Accept either a scoped population-pattern explanation or a technology pathway, provided the learner identifies a mechanism, one benefit and one trade-off. A national percentage must not be presented as class data or a proven cause. A convenience technology must not be claimed to guarantee safety. Useful evidence includes a properly annotated data card or source-to-eating system map.

6. Health and guidance

6. Health and guidance. Expected evidence: Look for a current public source used for its correct purpose, a cautious design or food-choice recommendation and explicit professional boundaries. “Supports” or “can reduce risk” is appropriate when the mechanism is accurate. Do not accept diagnoses, treatment promises, public body commentary or invented individual nutrient targets. Evidence may build on either guidance or the condition mechanism task.

7. HelloEats concept — draft only

7. HelloEats concept — draft only. Expected evidence: Expect a fictional consumer/context, a concept sketch or description, evidence-based choices and unconfirmed formal conditions listed clearly. The concept should address variety, safety, practicality and trade-offs. Require the wording “formative draft” and acknowledgement that the teacher’s current notification controls formal task conditions. A completed draft is not proof of practical work or submission.

8. Final synthesis and evaluation

8. Final synthesis and evaluation. Expected evidence: Expect a concrete change in reasoning supported by two pieces of unit evidence, such as replacing a per-serve shortcut with a matched comparison or recognising a missing claim source. The student should explain why the evidence changed their view and set one realistic next step. There is no compulsory personal disclosure. Assess honest evaluation rather than a generic statement that all work was enjoyable.

Evidence tools and review

Page 39 — Read the right evidence

The nutrition panel reports selected nutrient values on stated bases. It cannot identify every relevant allergen or apply an individual management plan. Check the current ingredients and allergen declaration, then the date, storage directions and the person's plan. Reject a choice that fails the safety requirement even if its nutrient numbers look attractive.

Page 40 — Compare equal amounts

Cereal C: $80 \text{ mg} \times 40 \text{ g} \div 100 \text{ g} = 32 \text{ mg}$ sodium. The calculation gives sodium for the assumed 40 g amount. It cannot establish overall nutritional suitability, actual consumption, allergen safety, other nutrients or the eater's needs. Accept any accurate limitation and a correctly labelled calculation.

Page 41 — Connect the ideas

Read down the left column, then the right. The six questions retain their original "1." numbering and use the exact first question from sections 1.1, 2.1, 3.1, 4.1, 5.1 and 6.1. Refer to the first answer on each corresponding section answer page. Ask students to explain one changed answer using the reading.

Page 42 — Glossary and references

Check terms in the student's explanation rather than accepting copied definitions alone. A strong response can distinguish an observation from an inference, a food pattern from an individual judgement, and a criterion from a personal preference.

Photography and observation

The original generated photographs are illustrative. They support recognising ingredients and discussing organisation, variety and planning. They cannot establish exact nutrient amounts, approved portions, allergen safety, storage conditions or actual student performance. The grain in the bowl is described generically because its variety is not visually established.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 01.1 — Personal hygiene and readiness

Scene A sources include the handled phone and everyday bag (possible biological transfer), loose hair (physical transfer), and unwashed hands; allergen transfer is possible only if an allergen is present, not proved by the scene. Scene B supplies storage, hair restraint, washing/drying, clean apron and a cleared work zone as controls.

An acceptable chain is phone → hand → ready-to-eat food; storing the phone then washing and drying hands interrupts it. Bag → shared surface → hands/food and hair → food are other reasonable chains.

The five-point checklist should order personal reporting/preparation before checking the work area and method, connect every action to a hazard, and leave local school procedures for teacher confirmation. A clear labelled drawing is the paper evidence equivalent.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 01.2 — Contamination, temperature and food safety

Potential hazards: raw-chicken board—biological; spray cleaner—chemical; loose container fragment—physical; used yoghurt spoon near a milk-free dish—milk-allergen transfer. Classifications concern plausible routes, not proof that transfer occurred.

Controls must be located at the route: separation of raw and ready-to-eat equipment/food, effective cleaning, protection from chemicals/fragments, allergen-aware separation and teacher-directed action. Do not describe handling the fictional hazards as a live practical.

Timer A: $70 + 35 = 105$ minutes, under 2 hours—use or refrigerate. B: $90 + 75 = 165$ minutes, 2–4 hours—use immediately, do not refrigerate for later. C: $150 + 110 = 260$ minutes, more than 4 hours—discard. Time is cumulative between 5°C and 60°C. The source rule is not a recipe or cooling instruction.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 01.3 — Safe decisions and incident response

Four records identify an observed fact and plausible harm: spill/slip; hot tray/burn or collision; fragments/physical contamination or injury; damaged lead/electrical injury. The first action is to stop and prevent further exposure without taking an untrained action, then tell the teacher and follow directions.

Accept context-aware stop/isolate/tell order where it keeps people away and obtains help. Students must not handle a hot tray, fragments, electricity or injured people beyond their training.

A redesign might keep drinks away from a route, establish a teacher-approved tray landing area, or check equipment before use. Before/after sketches must show which pathway is interrupted and how the change could be checked.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 02.1 — Six nutrients and their roles

Defensible labels include bread—carbohydrate, protein and some minerals; chickpeas—carbohydrate, protein and some minerals; yoghurt—protein, carbohydrate, minerals and water; apple—carbohydrate, water and vitamins; egg—protein, fat and vitamins/minerals; spinach—water, vitamins and minerals. Exact profiles depend on the food and preparation. Fibre is important but is not one of the six nutrient labels in this task.

The six cards span grains, vegetables/legumes, fruit, dairy/alternatives and protein foods. Foods can serve energy, growth/repair and regulation roles together; do not force one-to-one labels.

Model complementary-pairing explanation: Wholegrain bread and chickpeas both contribute carbohydrate, so their functions overlap in supplying energy. Both can also contribute protein, while their fibre and micronutrient profiles add further variety. A pairing therefore needs several labels rather than one label for each food. Combining them with vegetables can broaden the ingredients and food-group variety in a meal. The precise contribution depends on the ingredients, portions and preparation, so a photograph does not establish a complete nutrient profile. Removing an entire food group without a suitable alternative can narrow nutrient sources and choices. This is a contribution explanation, not a prescribed meal.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 02.2 — Energy, growth and repair

The correct numeric outcome is “insufficient evidence”: no nutrient panels, component masses or planned portions are supplied for the two mixed lunches. Do not invent equal energy values or add per 100 g values for different components.

Option A plausibly offers wholegrain, protein foods, vegetables, dairy and fruit; Option B includes dairy/grain components and fruit purée but requires detailed composition and portions. A conditional preference is defensible only if linked to Alex’s context and declared gaps.

The changed context reduces eating time and removes refrigerator access. Students should reconsider portability and cold-chain evidence, not declare that an insulated bag automatically makes a food safe. Allow alternative conditional decisions with sound evidence and unresolved allergy/cultural/medical needs.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 02.3 — Digestion and nutrient absorption

Order: mouth → oesophagus → stomach → small intestine → large intestine → rectum/anus. The liver, gallbladder and pancreas assist digestion but are not a route the food travels through.

Mouth includes chewing and the beginning of chemical digestion; muscular movement carries material along; the stomach mixes and contributes chemical digestion; the small intestine receives further digestive secretions and is the main site of nutrient absorption. Large intestine absorbs water from remaining material; waste leaves via rectum/anus.

At this simplified level: carbohydrate to simple sugars and protein to amino acids, absorbed into blood; many fat products are transported via intestinal lymph before entering blood. Do not imply all nutrients need digestion or all absorbed fat takes an identical route. The small intestine has a large folded/villus surface, thin absorbing surfaces and transport networks. Corrections distinguish movement, digestion and absorption.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 03.1 — Ingredients, servings and nutrition panels

Pass 1: both contain milk, use-by 10 Sep 20 XX and refrigeration at/below 5°C. This excludes either as an assumed milk-free choice. The placeholder date is not a current safety judgement.

Pass 2: compare similar products per 100 g: A energy 300 kJ, protein 7.0 g, sugars 5.0 g, sodium 50 mg; B 400 kJ, 4.5 g, 13.0 g, 60 mg. A saturated fat 1.6 g is slightly higher than B 1.5 g; do not claim A is lower in every measure.

Pass 3: serves differ 170 g versus 125 g. Per tub A 510 kJ versus B 500 kJ; protein 11.9 g versus 5.6 g; sugars 8.5 g versus 16.3 g. A's no-added-sugar statement does not mean sugar-free. The panels cannot prove overall dietary adequacy, exact added-sugar quantities or suitability for everybody. Retain both annotated label pages.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 03.2 — Comparing foods fairly

Claim A: B has lower stated tub energy 500 versus 510 kJ but higher energy per 100 g 400 versus 300 kJ; different tub masses explain the reversal. Claim B: A has 5.0 g sugars per 100 g and 8.5 g per tub despite no added sugar. Claim C: both declare milk; protein cannot override allergen suitability. Claim D: one serve is 170 g for A and 125 g for B, so one serve is not a shared mass.

Worked whole-tub calculation: $A\ 300 \times 1.70 = 510\text{ kJ}$; $B\ 400 \times 1.25 = 500\text{ kJ}$. For sugars, $B\ 13.0 \times 1.25 = 16.25\text{ g}$, rounded to 16.3 g as displayed. Accept clearly identified calculations from rounded figures.

A referee decision must name the portable-lunch purpose, compare common quantities then planned portions, recognise cold storage and milk for both, and state remaining user/cost/portion evidence. No single number automatically determines the choice.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 03.3 — Marketing claims, ethics and health

First impressions may be feelings/inferences rather than evidence. “Made with real berries” has an ingredient basis, but the amount is 2% berry purée; the first ingredients and coating provide context. “Fuel your best” has no defined outcome. “Natural energy” is not defined or demonstrated as a special benefit.

Check two claims against the printed ingredient evidence and identify what is missing, such as portion, nutrient panel, total price, audience requirements or packaging impacts. No invented nutrient values.

An acceptable fairer message says the product contains wholegrain oats 42%, yoghurt-style coating 26% and berry purée 2%, avoids undefined performance promises and keeps allergen information available. A respectful rewrite need not use the exact same words as the model.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 04.1 — Active non-nutrients and food variety

Wholegrain bread/oats/chickpeas/pear can contribute fibre; fruit/vegetables and yoghurt can contribute water; different foods contribute food-group or colour variety. A label can apply more than once. Plain yoghurt is not ordinarily a fibre source. Water is a nutrient.

Repair 1: a smoothie can contribute nutrients from its ingredients, but the statement supplies no evidence of overnight detoxification. Repair 2: chia can contribute fibre and other nutrients as part of an eating pattern; one spoonful cannot guarantee prevention of heart disease. Merely inserting “may” before an unsupported detox claim is not an adequate repair.

Snack redesign should name a purpose, add a meaningful contribution, remain realistic for its stated context, avoid moral labels and leave allergy or practical suitability for confirmation. Six classified foods and two distinct repairs are required.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 04.2 — Nutrition across the lifecycle

Accept any two of the four cards. Stated priorities may include Jordan's short break/portability/familiar savoury food; Ari's softer easy-to-hold food and social meal; Sam's separate components/smaller easy-to-hold wrap; Lee's short break and available fridge/microwave.

Adapt ingredients, texture, presentation, portability or preparation only in response to stated facts. Age does not establish an illness, appetite, chewing ability or exact nutrient requirement. Do not invent a portion prescription.

Each adaptation needs a reason, a practical check and an explicit unknown such as allergy, cultural need, budget or reliable storage. A peer review should distinguish a confirmed fact from an assumption; current school practical directions still apply.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 04.3 — Personal, social, cultural and digital influences

Visible cues include a sponsored sales purpose, detox claim, "zero guilt" emotional framing, 20% discount and 10-minute scarcity pressure, plus tap-to-order convenience. Influence categories may overlap; explain arrows rather than guess private identity or household facts.

A claim check identifies the sponsor's persuasive purpose, absence of detox evidence and missing ingredients, serving size and total price. A discount is not evidence of need or overall value.

A defensible pause-check-decide response could pause, seek complete price/ingredient/evidence information, then delay or choose another option. Different outcomes are acceptable if justified and non-judgemental.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 05.1 — Australian food-consumption patterns

Annotation: ABS; Australian children aged 2–17; consumption year 2023; percentage meeting recommendations; vegetables/legumes/beans 2.1%, fruit 50.7%; release 10 July 2026. These are national population estimates.

Claim A is supported. B overgeneralises to the school and changes meeting a recommendation to eating any vegetables. C is unsupported causation. D overgeneralises not meeting a recommendation to eating no fruit. Repair B/D with the actual population, measure and limits.

The data motivates investigation of possible barriers such as access, affordability, time or acceptability; it establishes none as the cause. A proportionate response might trial an appealing affordable vegetable option and measure uptake/feedback without diagnosing or judging individuals.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 05.2 — Convenience, delivery and food technology

Sequence both options through sourcing, preparation, packaging, transport, handover and storage/eating. Technology placement must make sense: refrigeration in storage; microwave in the home preparation; app at ordering; packaging during packing/transport. Distinguish observed case facts from assumptions.

Home-assembled time is about 12 minutes; delivered estimate 25–40 minutes may miss the 30-minute aim. Home case supplies ingredient/storage information; delivered ingredients/allergen controls/portion/arrival temperature are unconfirmed. Budget/preferences/allergies remain unknown for both.

Potential control points include confirming ingredients/allergens and maintaining safe time-temperature handling. An acceptable redesign improves a stated stage while preserving convenience, such as clearer ingredient information or a better-verified delivery window; it must not invent a safety guarantee or unsupported price.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 05.3 — Designing responsible food solutions

Constraints include 1 pm cold lunch, available insulated bag, ingredient-cost target ≤ \$5 and home assembly ≤ 15 minutes. Allergy/cultural/medical requirements remain unconfirmed. Criteria may include variety, appeal, portability, time, cost and packaging; keep constraints distinct.

Weights 1–3 are an explicitly original scaffold. Accept different justified weights. Scores need actual evidence; concepts alone do not supply measured prices, preparation times or nutrient profiles. “?” is the correct treatment of unknown evidence. An allergy/safety gap cannot be cancelled by a high numeric score.

Under the eight-minute stress test, identify which evidence must be gathered again. Recommend conditionally and describe one observable prototype test, for example a timed assembly trial under approved conditions or a packaging test. No recipe or practical approval is created by this worksheet.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 06.1 — Under- and overnutrition

Case 1: recorded skipped breakfasts and limited lunches linked to household shortage plus reported tiredness; a possible mechanism is insufficient energy or nutrient intake, but tiredness has many causes and no diagnosis is supplied. Respectful support could address food access privately with an appropriate adult/help pathway.

Case 2: repeated takeaway/sweet drinks during late work are observations; a possible high-energy pattern depends on actual portions and needs. Activity, sleep, medicines and health are unknown. Avoid equating one food, body size or household choice with a diagnosis.

Case 3: narrow recorded variety can make adequate fibre or some micronutrient intake harder; actual needs and medical information remain unknown. Suggest inclusive access to varied choices and an appropriate private support route.

Context cards are possibilities only. Reject diagnosis from appearance, supplement prescription, public disclosure and cure guarantees. Three separate evidence–mechanism–support triads plus an explicit boundary explanation are required.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

COMPANION / MODEL REASONING

Applied activity solutions

Activity 06.2 — Nutrition-related conditions and prevention

Accept one accurate source-bounded mechanism: iron participates in haemoglobin and oxygen transport; cardiovascular risk reflects interacting factors; calcium/vitaminD and other factors contribute to bone health; type 2 diabetes involves impaired insulin action/regulation with multiple influences. No single symptom proves a condition.

Factors: smoking is modifiable; physical activity opportunities and food availability have environmental and modifiable dimensions; age/family history are non-modifiable; healthcare access is environmental; one tiredness report needs more context. Categories may overlap when explained.

Repair A: tiredness has many possible causes; anaemia requires appropriate professional assessment. Repair B: calcium-rich foods can support bone health within an eating pattern, but no food guarantees prevention. Repair C: type 2 diabetes has interacting genetic, age, environmental and behavioural influences; blame is inaccurate and stigmatising.

An inclusive response might improve access to varied appropriate food or activity opportunities without singling anyone out. A private trusted-adult/qualified-health-professional pathway is appropriate; student diagnosis, treatment or public disclosure is not.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

Activity 06.3 — Australian guidance and the HelloEats challenge

Robin's stated portability, short break and preferences can become criteria; budget, allergies/cultural/portion requirements and cold-storage arrangements remain confirmation points. Food-group variety is considered across a day; do not impose an identical plate.

Use complete yoghurtA/B data. Per 100 gA has lower energy and sugars, more protein, and slightly higher saturated fat. Tub amounts differ; planned portion matters. Both contain milk and need refrigeration. No Health Star Rating is supplied.

The concept needs annotations for food groups, appeal, honest cost/packaging evidence and safety/allergen confirmation; it remains a concept, not an approved recipe. Accept different defensible concepts.

The final status must explicitly read formative evidence/not submitted, with formal task title/number/date/marks/criteria/submission/practical conditions traced only to the current teacher-issued notification. A site export or completed workbook is not proof of submission.

Accept source-supported alternatives with explicit evidence and limits; do not require identical model wording.

TEACHER GUIDANCE / EXTENSION

Optional reasoning puzzles

1. Safety decision chain

A defensible sequence is identify the hazard → stop unsafe work → protect people where safe → tell the teacher → follow the approved response → record/review. Immediate recognition, stopping and warning can overlap; do not demand a slow rigid order during danger. The student should explain that controls reduce immediate harm, trained people direct the response and later review reduces recurrence. Never delay stopping an unsafe action simply to complete a written identification step. Accept reordered protective/communication actions when the scenario justifies them and the student remains within training.

2. Nutrient function match

One example is chickpeas → protein → amino acids used in growth, repair and body components. A second is yoghurt → calcium → contribution to bones and teeth. The limit must prevent overclaiming: one food does not meet every nutrient requirement, prove individual adequacy or guarantee disease prevention. Accept other accurate food–nutrient–function chains, including plant sources. Do not treat the selected food as containing only one nutrient; its additional contributions may be acknowledged.

3. Label detective

Serving size answers “what amount is used for the per-serve figures?” Per-100-g values answer “how do similar foods compare at the same amount?” Ingredient order answers “which ingredients contributed more ingoing weight?” Relevant claims answer a narrower question that requires matching label evidence, such as an identified nutrient feature. After standardisation, return to the realistic portion and the eater’s purpose. A full answer recognises allergen/storage needs and that ingredient order does not reveal all exact quantities. The fictional yoghurt pair can provide complete paper evidence.

4. Influence matrix

Accept six justified examples such as taste preference (personal), a friend’s suggestion (social), a family celebration (cultural), a spending limit (economic), packaging concerns (environmental) and a sponsored discount post (digital). Categories overlap. Two explained interactions might show a discount reinforcing budget appeal, and a family preference conflicting with available time or a platform’s limited options. Require arrows or sentences that explain the relationship rather than six isolated labels. Alternative classifications are valid when the learner explains them without stereotyping or blame.

5. Guidance-to-design challenge

Example: for a fictional learner needing a portable lunch, add a suitable vegetable and legume component to broaden food-group variety while checking the available storage and time constraint. Evidence should identify the relevant Australian guidance and, where used, a fair label comparison. A trade-off might be extra preparation against convenience, or container protection against material use. The concept remains formative: no recipe, allergy clearance, assessment completion or formal submission is implied. Accept varied designs linked to a named user, evidence and one actual constraint.

SOURCE NOTES

Course sources and authority

Course site: <https://stevencowell.github.io/Year-9-Food-Selection-and-Health/>

Snapshot revision: c5e33e886b2ccdbf79cceac06f4cd6957240d247 (3 September 2026). Live source checks were completed for this build. The old Food-Selection-and-Health route is obsolete.

Modules 1–6 are at /modules/module-01.html through /modules/module-06.html. Their section anchors m01-s1 through m06-s3 identify the 18 learning sections. The matching /activities.html, /folio.html and /assessment.html supply applied evidence, folio stages and assessment orientation.

Source decisions

Water is consistently treated as one of the six nutrient groups. A sentence in the source's section 4.1 groups it confusingly with non-nutrient components; the workbook resolves that wording using the source's section 2.1 definition. Fibre and plant compounds remain distinct from the six nutrient groups.

Some module handoffs use sub-evidence names that differ from the eight actual folio stages. Retain those named pieces of work and file them under the nearest broader stage. They do not create extra required online folio stages.

Original source questions and option order are retained. Additional missing stimulus cards in the companion are explicitly labelled fictional teaching aids. They preserve the task's purpose and do not assert local school procedures or real product claims.

Assessment orientation

HelloEats is a draft nutritious ready-to-eat meal concept for a named consumer. The teacher's current notification controls formal requirements. This package does not reproduce an unverified task sheet or invent marks, due dates, recipe approval or submission proof.

REFERENCE / 1 OF 3

Public reading sources

NSW Food Authority — Food safety at home
<https://www.foodauthority.nsw.gov.au/consumer/food-at-home>

Food Standards Australia New Zealand — Food safety
<https://www.foodstandards.gov.au/consumer/safety>

NSW Curriculum — Food Technology 7–10 (2025) outcomes
<https://curriculum.nsw.edu.au/learning-areas/tas/food-technology-7-10-2025/outcomes>

Food Standards Australia New Zealand — 2-hour / 4-hour rule
<https://www.foodstandards.gov.au/business/food-safety/2-hour-4-hour-rule>

SafeWork NSW — Hazards A–Z
<https://www.safework.nsw.gov.au/hazards-a-z>

Healthdirect Australia — Burns and scalds
<https://www.healthdirect.gov.au/burns-and-scalds>

Australian Government Eat for Health — Nutrient Reference Values
<https://www.eatforhealth.gov.au/nutrient-reference-values>

REFERENCE / 2 OF 3

Public reading sources

Australian Guide to Healthy Eating

<https://www.eatforhealth.gov.au/guidelines/australian-guide-healthy-eating>

Australian Government Eat for Health — Australian Dietary Guidelines

<https://www.eatforhealth.gov.au/guidelines>

Australian Government Eat for Health — Healthy eating throughout life

<https://www.eatforhealth.gov.au/eating-well/healthy-eating-throughout-all-life>

Food Standards Australia New Zealand — Nutrition information panels

<https://www.foodstandards.gov.au/consumer/labelling/panels>

Healthdirect Australia — Digestive system

<https://www.healthdirect.gov.au/digestive-system>

TED-Ed — How your digestive system works

<https://www.youtube.com/watch?v=Og5xAdC8EUI>

Food Standards Australia New Zealand — Ingredient lists

<https://www.foodstandards.gov.au/consumer/labelling/ingredients>

REFERENCE / 3 OF 3

Public reading sources

Food Standards Australia New Zealand — Allergen labelling
<https://www.foodstandards.gov.au/consumer/labelling/allergen-labelling>

Health Star Rating — How to use health stars
<https://www.healthstarrating.gov.au/>

Food Standards Australia New Zealand — Nutrition and health claims
<https://www.foodstandards.gov.au/consumer/labelling/nutrition>

Australian Competition and Consumer Commission — False or misleading claims
<https://www.accc.gov.au/consumers/advertising-and-promotions/false-or-misleading-claims>

Australian Government — Eat for Health
<https://www.eatforhealth.gov.au/>

Australian Institute of Health and Welfare — Food and nutrition overview
<https://www.aihw.gov.au/reports-data/behaviours-risk-factors/food-nutrition/overview>

Australian Bureau of Statistics — Consumption of food groups from the Australian Dietary Guidelines, 2023
<https://www.abs.gov.au/statistics/health/food-and-nutrition/consumption-food-groups-australian-dietary-guidelines/latest-release>